3. Project Design Phase

Architecture, modules, interface and data flow

Architecture

The documented design uses a FastAPI backend connected to a simple HTML/CSS frontend. The backend connects the appropriate educational module to either Gemini-based cloud inference or the local LaMini-Flan-T5 model.

Model Allocation

- Gemini 1.5 Pro via API: Q&A;, summarization, quiz generation and learning paths.
- LaMini-Flan-T5-783M: concept explanation.

Folder Architecture

- main.py — FastAPI application.
- explanation_module.py — concept explanation logic.
- qna.py — question answering.
- quiz_module.py — quiz generation.
- summary_module.py — summarization.
- learning_path.py — learning recommendations.
- templates/index.html — HTML frontend.
- static/style.css — styling.
- requirements.txt — Python dependencies.

API Design

- /qa
- /explain
- /quiz
- /summarize
- /learn/recommendations

Frontend Design

The interface contains a task dropdown, a text area, a submit button, responsive styling, and a result container. The documented workflow sends a POST request to the backend and displays the result below the input area.

Quiz Output Design

The quiz module is designed to generate three MCQs with four options each and a correct answer, using structured JSON for integration.